NeuroAgent

A clinical collaboration proposal: validating an evidence-driven neurological diagnosis benchmark

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A new way of testing medical AI

Most research today tests medical AI with question answering. Real clinical work is not like that.

Today: question answering

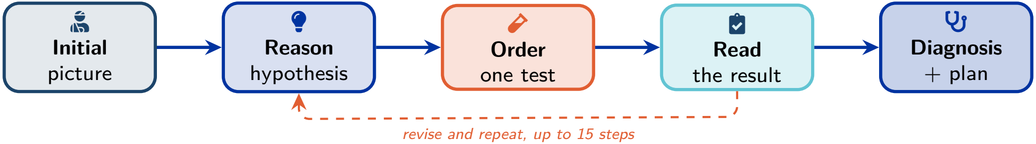
- Multiple-choice exams
- Every clue is already in the question
- One correct option to pick
- Rewards recall, not reasoning

Real clinical practice

- The patient arrives with a complaint, not options
- You decide which investigations to order
- Results come back over time
- Cost, risk and safety all matter

We introduce a **new paradigm**: test the AI on the whole diagnostic workup, the way a clinician actually reasons.

Project overview



600 realistic neurology cases across 20 conditions.

What we measure

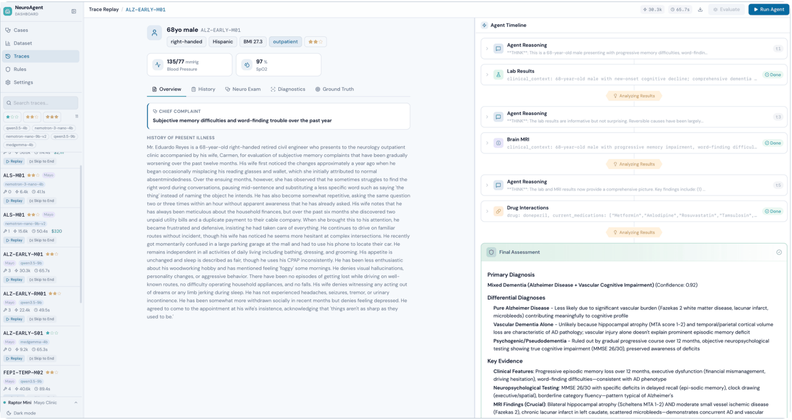
- Right test, right order, sensible cost
- Sound reasoning, not memorised facts
- A workup a clinician can follow

Why it is new

- Today's AI exams hand over every clue
- Tests the real skill: what to investigate
- Each case checked by **2+ clinicians**

The agent in action

The goal: a transparent workup, ending in a diagnosis backed by evidence.



The dataset

Cases

600

20 conditions

Per condition

30

exactly, fully balanced

Follow-up results

3,800

triggered by actions

How each case is built

- Patient core: history, exam, vitals
- Evidence hidden behind tool calls
- Ground truth: best path, red herrings, unsafe actions

Spanning

- 3 difficulty tiers: straightforward to puzzle
- 396 synthetic + 204 real-report-seeded
- 3 settings: emergency / inpatient / outpatient

Results are cached in two tiers: ***initial*** *findings when a test is first ordered, and* ***follow-ups*** *unlocked by a specific next action.*

How the cases were generated

Two pipelines, one schema: real cases for fidelity, synthetic cases for scale.

Seeded from real case reports

- Source: **MedCaseReasoning**, clinician-written case reports
- Rewritten by **Claude Opus 4.8 + Sonnet 4.6**: findings hidden in tool outputs

204 cases

Fully synthetic, for scale

- Built from per-condition clinical schemas, no source report
- Scale needed to **fine-tune local models**
- Same schema and ground truth as the seeded cases

396 cases

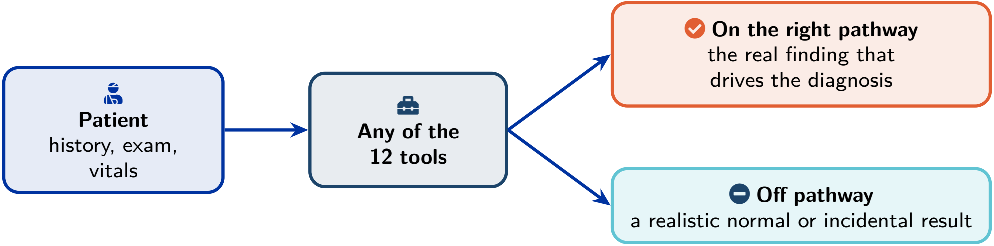
The 12 diagnostic tools

Tool	What it returns
Brain MRI	report with region-tagged findings
EEG	background, epileptiform features, focality
ECG	rhythm, intervals, signs of ischaemia
Labs	panels with reference ranges + abnormality flags
CSF (lumbar puncture)	cells, protein, glucose, NfL, oligoclonal bands
CT scan	non-contrast / contrast, head / spine
Echocardiogram	chambers, valves, ejection fraction, thrombus
Cardiac monitoring	telemetry / Holter / loop recorder
Advanced imaging	MRA / MRV / CTA / PET / DaT-SPECT
Specialised tests	EMG/NCS, biopsy, autoimmune panels
Literature search	guideline and paper snippets
Drug interactions	interaction profile + contraindications

Each returns a structured, typed report with a cost; any tool, any time.

Every patient is a complete picture

Each case pre-computes a whole patient, so the agent can order **any** of the 12 tools and always get a realistic answer.



Never a dead end: every tool answers. But each call costs money and a turn, so ordering the wrong test is penalised.

A worked example: wake-up ischemic stroke

Turn 1: what the agent sees

- 58 F: woke with slurred speech, clumsy hand
- Last known well 9 h ago: window unclear
- Left face/arm weak, sensory loss, neglect

What the right workup reveals

- CT first: rule out bleed; glucose: no hypoglycemia
- MRI: fresh infarct, deep right white matter
- ECG, echo, carotids, Holter: hunt the cause

The trap, and an off-pathway test

"Chronic, not acute" tempts an early stop, yet the cause hunt is still owed. An off-pathway EEG reads **normal**: wasted.

Ground truth: acute ischemic stroke, right corona radiata (ICD-10 I63.541).

How we evaluate performance

Automatic metrics, per case

- **Diagnosis:** ICD top-1 / top-3 match
- **Tool selection:** precision, recall, F1 + coverage
- **Safety:** contraindicated, harmful, wrong-order calls
- **Cost:** $ vs optimal; useless or repeated calls

AI reasoning judge, 0–5 each

- Diagnostic accuracy; evidence found + integrated
- Differential reasoning; uncertainty calibration
- Tool efficiency, safety, red-herring handling
- One **composite reasoning score**

Clinician validation: the gold standard

$\geq$2 clinicians per case: field-level approval + inter-rater agreement, the human ceiling we calibrate against.

Goals for the medical collaboration

1. Validate the cases

- Read each patient as you would in clinic
- Approve, flag, or correct field by field
- Drop cases that are not good enough

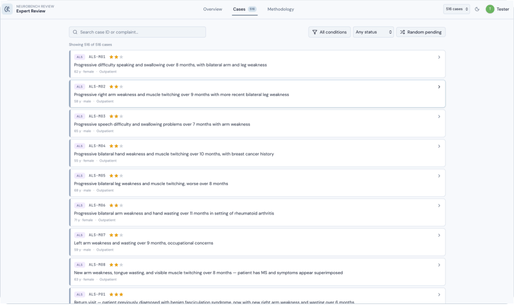
2. Validate our scoring

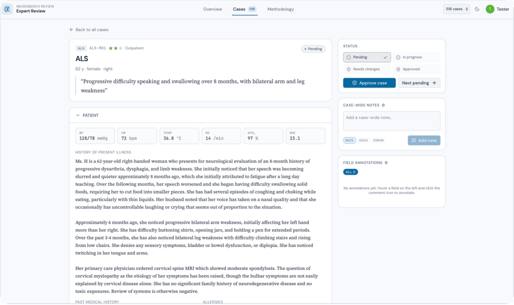
- Score a sample of agent runs yourself
- Compare against the AI reasoning judge
- Confirm scores track clinical quality

How the review works

- $\geq$**2 independent reviewers** per case
- Separate login per reviewer: blind, no shared bias
- Agreement stats $\rightarrow$ fix or drop weak cases $\rightarrow$ higher quality

The annotation platform



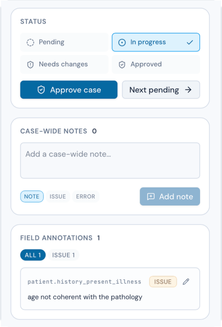


Browse cases: condition, difficulty, status

One screen per patient: vitals, history, full workup

review.andreaprotani.com

Reviewing a case



Flag any field: note, issue, or error

Per-case status, every annotation logged

Approve, request changes, or move on: progress tracked per reviewer.

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